MiraFood nutrition formulas

Reference for how patient profile targets, macros, water, meal nutrition, and health scores are calculated.

Authoritative engine: `server/src/modules/consumers/nutrition-calculation-engine.ts`
Version: `2026-07-clinical-v1` (`NCE_VERSION`)
Related: `profile-targets.util.ts` , `dashboard.util.ts` , `nutrient-score.util.ts` , `meals/nutrition.util.ts`

Mobile/web may show **provisional previews** during onboarding (`mobile/src/utils/nutrition.ts`). Stored clinical targets always come from the **server NCE**.

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1. Patient profile inputs

Collected on onboarding (patient)

Field	Use
Date of birth → age	BMR age term; pediatric vs adult path
Sex	Mifflin / Schofield sex constants
Height (cm), weight (kg)	BMR, BMI, protein, water
Activity level	TDEE multiplier (adults / pregnancy / lactation)
Goal (+ optional pace)	Calorie surplus/deficit **after coach confirmation**
Meals per day	Consistency component of health score
Preferences / allergies	Not used in calorie/macro math

Coach assessment (overrides / clinical)

Coach-verified values override patient basics when present (`mergedCalculationProfile`):

- Verified DOB/age, sex, height, weight
- Pregnancy / trimester / babies / pre-pregnancy weight
- Lactation
- Conditions, fluid restriction
- Goal pace; optional `coachAllowsProtectedWeightLoss`

Target status

Status	Meaning
`unavailable`	Missing age / height / weight / activity / goal
`provisional`	Patient complete, or clinical flags still open
`confirmed`	Coach confirmed assessment and no blocking flags

Rule: Goal deficits/surpluses (`goalAdjustmentKcal`) apply only when `assessmentStatus === "confirmed"` . Until then, calorie target = maintenance TDEE.

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2. BMI

[\mathrm{BMI} = \frac{\mathrm{weight_kg}}{(\mathrm{height_cm}/100)^2}]

Stored rounded to **1 decimal** (`Math.round(bmi * 10) / 10`).

Flag: BMI **> 40** → `bmi_over_40_coach_review` (still uses actual weight in Mifflin–St Jeor).

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3. BMR (basal metabolic rate)

Adults (age ≥ 18) — Mifflin–St Jeor

[\mathrm{base} = 10,W + 6.25,H - 5,A]

Sex	BMR	`equationUsed`
Male	(\mathrm{base} + 5)	`mifflin_st_jeor_male`
Female	(\mathrm{base} - 161)	`mifflin_st_jeor_female`
Other / prefer not to say / null	Average of male & female	`mifflin_st_jeor_sex_average`

Where (W) = kg, (H) = cm, (A) = years.

Rounded: `Math.round(bmr)`.

Pediatrics (age < 18) — Schofield (weight-based)

Age	Boys	Girls
< 3	(59.512,W - 30.4)	(58.317,W - 31.05)
3–9	(22.706,W + 504.3)	(20.315,W + 485.9)
10–17	(17.686,W + 658.2)	(13.384,W + 692.6)

Unknown sex → average of boy and girl equations.
Pediatric **TDEE = BMR** (no activity multiplier until a validated pediatric reference is approved).

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4. Activity multipliers (TDEE)

Activity level	Multiplier
`sedentary`	1.2
`lightly_active`	1.375
`moderately_active`	1.55
`very_active`	1.725
`extremely_active`	1.9

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5. TDEE by population

Healthy adult

[\mathrm{TDEE} = \mathrm{BMR} \times \mathrm{activity_multiplier}]

Pregnancy

- BMR from Mifflin using **pre-pregnancy weight** when available (else current weight + warning).
- Then:

[\mathrm{TDEE} = \mathrm{BMR} \times \mathrm{activity_multiplier} + \mathrm{pregnancy_addition}]

Case	Addition (kcal)
Trimester 1 (or missing)	0
Singleton trimester 2	+340
Singleton trimester 3	+452
Twins, trimester 2 or 3	+685

Lactation

$$ \mathrm{TDEE} = \mathrm{BMR} \times \mathrm{activity_multiplier} + 500 $$

(BMR uses **current** weight.)

Pediatric

$$ \mathrm{TDEE} = \mathrm{BMR}_{\mathrm{Schofield}} $$

Rounded: `Math.round(tdee)`.

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6. Goal calorie adjustment

Applied **only if** coach assessment is `confirmed`:

Weight loss (`lose_weight`)

Pace	Adjustment
`slow` (default)	−500 kcal
`moderate`	−625 kcal
`aggressive`	−750 kcal

Muscle gain (`gain_muscle`)

Pace	Adjustment
`slow` (default)	+300 kcal
`moderate`	+400 kcal
`aggressive`	+500 kcal

Maintain / improve diet quality

$$ \mathrm{adjustment} = 0 $$

Protected populations

Pregnancy, lactation, and pediatric **never** get an automatic deficit unless `coachAllowsProtectedWeightLoss` is true. Deficit is forced to `0` and flagged `automatic_weight_loss_blocked`.

Calorie target

$$ \mathrm{calorieTarget} = \max(0,\ \mathrm{round}(\mathrm{TDEE} + \mathrm{adjustment})) $$

Stored as `macroTargets.calories`.

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7. Macro targets

Using calorie target $C = \mathrm{calorieTarget}$ and weight W (kg):

Protein

$$\mathrm{protein_g} = \begin{cases} \mathrm{round}(2.0 \times W) & \text{if goal = gain_muscle} \\ \mathrm{round}(1.6 \times W) & \text{otherwise} \end{cases}$$

Energy: 4 kcal/g.

Fat

$$\mathrm{fat_g} = \mathrm{round}\left(\frac{0.28 \times C}{9}\right)$$

(~28% of calories from fat; 9 kcal/g.)

Carbohydrate (remainder)

$$\mathrm{carbs_g} = \max\left(0,\ \mathrm{round}\left(\frac{C - 4\cdot\mathrm{protein_g} - 9\cdot\mathrm{fat_g}}{4}\right)\right)$$

Fiber

Sex	Fiber (g/day)
Male	38
Female / other / unset	25

Worked check (energy identity)

$$C \approx 4\cdot P + 4\cdot \mathrm{carbs} + 9\cdot F$$

(Carbs absorb rounding remainder.)

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8. Water target

$$\mathrm{waterTargetMl} = \mathrm{round}(W \times 35)$$

(~35 ml per kg). Flagged for clinical review if fluid restriction / kidney / heart disease — never override a prescribed restriction automatically.

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9. Meal / ingredient nutrition (portions)

Scale by weight

If nutrition is known for weight w_0, new weight w_1:

$$\mathrm{ratio} = \frac{w_1}{\max(w_0, 1)},\quad \mathrm{nutrient}' = \mathrm{nutrient} \times \mathrm{ratio}$$

Calories/sodium typically **integer-rounded**; macros to **2 decimal places**.

From per-100 g composition

$$\mathrm{factor} = \frac{w}{100},\quad \mathrm{nutrient}(w) = \mathrm{nutrient}_{100\mathrm{g}} \times \mathrm{factor}$$

Meal totals

Sum of ingredient calories / P / C / F / fiber (and micronutrients when present).

What counts on the diary / dashboard

Only **coach-approved** meals for the calendar day contribute to consumed macros, calories, and health score. Approved totals prefer coach `totalNutrition` / reviewed items when present.

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10. Daily health score

Components are each clamped to **0–100**.

Ratio score (calories & macros & nutrients)

$$ r = \frac{\mathrm{actual}}{\mathrm{target}} $$

- If $r \le 1$: score $= 100\,r$
- If $r > 1$: score $= 100 - 35\,(r - 1)$ (overshoot penalty), then clamp

Nutrient adequacy (30% of total)

- Always scores **fiber** vs fiber target.
- Also scores these micros **when present** on meal items:

Key	Daily target
`ironMg`	18
`vitaminCMg`	90
`calciumMg`	1000
`vitaminDIu`	600
`potassiumMg`	3500

Average of available part scores → `measuredScore`.
Coverage $d = \mathrm{available}/6$ (fiber + 5 micros).

$$ \mathrm{nutrientScore} = \mathrm{measuredScore} \times (0.5 + 0.5\,d) $$

(Missing micros cannot look like perfect adequacy.)

Macro score (25%)

Average of protein, carbs, fat ratio scores vs targets.

Calorie score (20%)

Ratio score of consumed vs calorie target.

Consistency (15%)

$$ \mathrm{consistency} = \min\left(100,\ \frac{N_{\mathrm{approved\ meals\ today}}}{\max(1,\ \mathrm{mealsPerDay})} \times 100\right) $$

Variety (10%)

Distinct food labels today (else distinct meal names), target **5**:

$$ \mathrm{variety} = \min\left(100,\ \frac{\mathrm{distinct}}{5} \times 100\right) $$

Combined score

[\begin{aligned} \mathrm{healthScore} =\ &\mathrm{round}(0.30,\mathrm{nutrient}

- 0.25,\mathrm{macro}
- 0.20,\mathrm{calorie} \ &\quad + 0.15,\mathrm{consistency}
- 0.10,\mathrm{variety}) \end{aligned}]

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11. Client preview vs server (important)

	Mobile onboarding preview	Server NCE (stored)
Goal adjustment	Applied immediately (−500 / +300 defaults)	Only after coach **confirm**
Calorie floor	`max(1200, TDEE + adj)`	`max(0, TDEE + adj)`
Pregnancy / lactation / Schofield	Not in mobile util	Full NCE paths
Pace (slow/mod/aggressive)	Not in mobile util	Full NCE table

UI copy should treat onboarding numbers as **estimates** until coach confirmation when clinical workflow is active.

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12. Example (confirmed healthy adult male)

Input: 30 y, male, 180 cm, 80 kg, moderately active, maintain, confirmed.

[\begin{aligned} \mathrm{BMR} &= 10(80) + 6.25(180) - 5(30) + 5 = 1780 \ \mathrm{TDEE} &= 1780 \times 1.55 = 2759 \ C &= 2759 \ P &= \mathrm{round}(1.6 \times 80) = 128\ \mathrm{g} \ F &= \mathrm{round}(2759 \times 0.28 / 9) = 86\ \mathrm{g} \ \mathrm{carbs} &= \mathrm{round}((2759 - 128\cdot4 - 86\cdot9)/4) = 362\ \mathrm{g} \ \mathrm{fiber} &= 38\ \mathrm{g} \ \mathrm{water} &= \mathrm{round}(80 \times 35) = 2800\ \mathrm{ml} \ \mathrm{BMI} &= 24.7 \end{aligned}]

(Matches `nutrition-calculation-engine.test.ts`.)

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13. Code map

Concern	File
BMR / TDEE / macros / water / flags	`server/.../nutrition-calculation-engine.ts`
Merge coach + patient → calculate	`server/.../profile-targets.util.ts`
Dashboard consumed + health score	`server/.../dashboard.util.ts`
Micronutrient adequacy	`server/.../nutrient-score.util.ts`
Portion scale / meal sum	`server/.../meals/nutrition.util.ts`
Onboarding preview only	`mobile/src/utils/nutrition.ts`
High-level clinical policy	`docs/clinical-nutrition-engine.md`

Disclaimer

These formulas support educational coaching and logging. MiraFood is **not** a medical device and does not replace individualized clinical nutrition care. Protected populations and high-risk conditions require coach review before targets are treated as confirmed.